## Test Case / Original Request

Run test case PTC-004 using k6, monitor it, analyze the results, and generate the final PDF report with links to the native JMeter report and raw results.

## REAL Execution Outcome

Execution status: COMPLETED
Runner status: completed
Outcome details: REAL k6 execution completed. Threshold breaches, if any, are retained as performance findings.
Thresholds passed: True

## Executive Summary

Application: Unknown
Scenario: Filtered Feedback API — Load Check
Job ID: JOB-74e8bfd0b802423fbf98b58f157d7dcd
Overall Result: FAIL

This run of the Filtered Feedback API — Load Check scenario experienced a critical performance failure, with error rates and max response times dramatically breaching thresholds. Infrastructure monitoring recorded a warning, driven by high memory usage on the load generator host. No relevant security results or historical analysis context were available. Action is recommended on both error handling and infrastructure alerting.

## Execution Profile and Artifacts

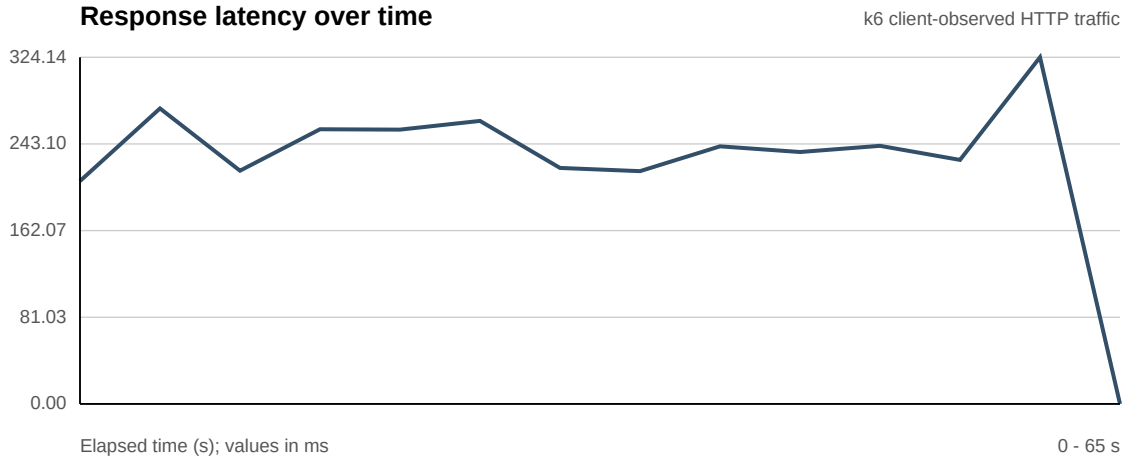
Execution mode: REAL
Workload profile:
- tool: k6
- target_url: https://jsonplaceholder.typicode.com
- endpoint: /comments?postId=1
- method: GET
- users: 20
- max_rps: 12
- duration_seconds: 60
- ramp_up_seconds: 5
- expected_status_code: 200
- p95_threshold_ms: 5000
Artifact directory: src/generated_scripts/k6 native reports/JOB-74e8bfd0b802423fbf98b58f157d7dcd
Artifact references:
- aligned_time_series: src/generated_scripts/k6 native reports/JOB-74e8bfd0b802423fbf98b58f157d7dcd/k6_PTC_004_Filtered_Feedback_API_Load_Check_aligned_time_series.csv
- generated_script: src/generated_scripts/k6 native reports/JOB-74e8bfd0b802423fbf98b58f157d7dcd/k6_PTC_004_Filtered_Feedback_API_Load_Check.js
- k6_summary: src/generated_scripts/k6 native reports/JOB-74e8bfd0b802423fbf98b58f157d7dcd/k6_PTC_004_Filtered_Feedback_API_Load_Check_summary.json
- measurement_metadata: src/generated_scripts/k6 native reports/JOB-74e8bfd0b802423fbf98b58f157d7dcd/k6_PTC_004_Filtered_Feedback_API_Load_Check_measurement_metadata.json
- monitoring_samples: src/generated_scripts/k6 native reports/JOB-74e8bfd0b802423fbf98b58f157d7dcd/k6_PTC_004_Filtered_Feedback_API_Load_Check_runtime_samples.csv

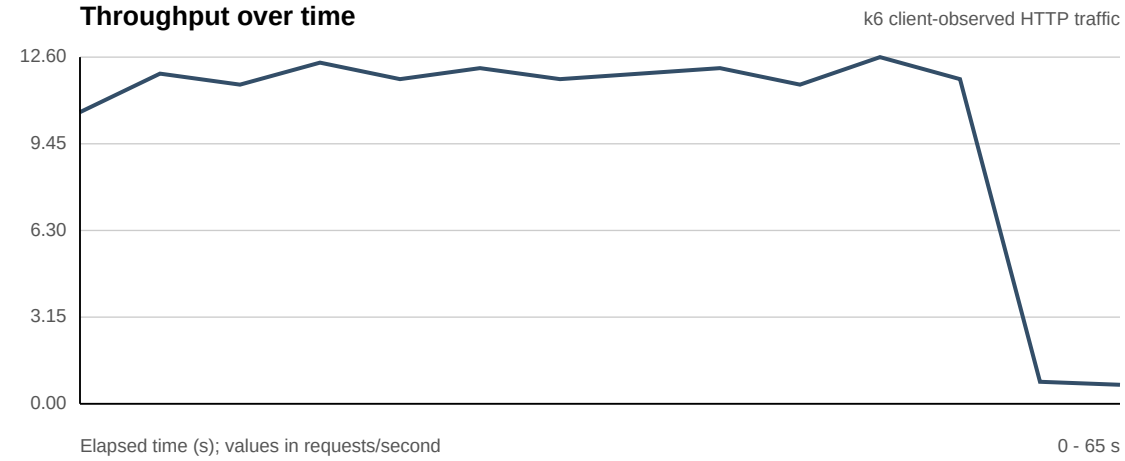
- Native k6 HTML report
- Native report + raw results ZIP
- normalized_metrics: src/generated_scripts/k6 native reports/JOB-74e8bfd0b802423fbf98b58f157d7dcd/k6_PTC_004_Filtered_Feedback_API_Load_Check_normalized_metrics.json
- Raw k6 results (JSON)
- runner_log: src/generated_scripts/k6 native reports/JOB-74e8bfd0b802423fbf98b58f157d7dcd/k6_PTC_004_Filtered_Feedback_API_Load_Check_runner.log
- time_series: src/generated_scripts/k6 native reports/JOB-74e8bfd0b802423fbf98b58f157d7dcd/k6_PTC_004_Filtered_Feedback_API_Load_Check_time_series.csv

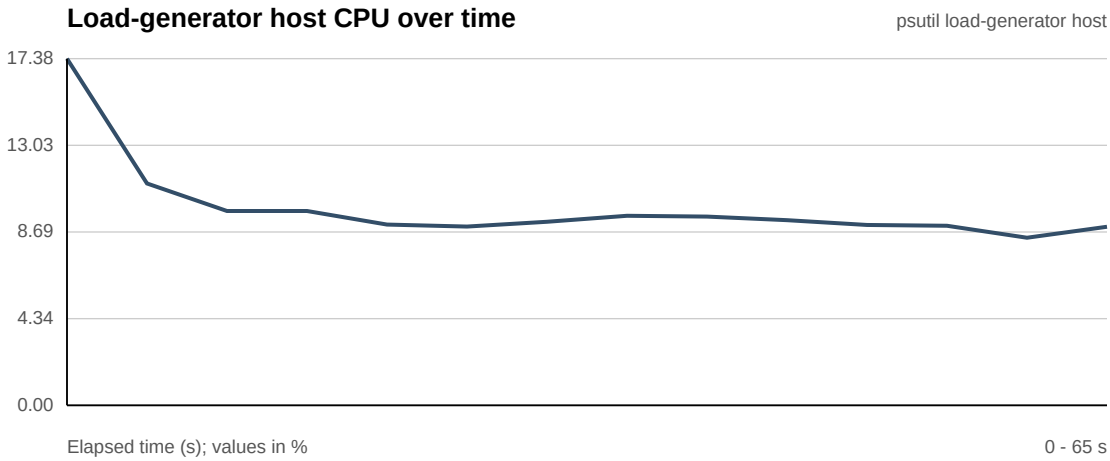
The interactive native HTML dashboard is linked as a separate artifact and is not embedded in this PDF.

## Time-Series Charts (REAL Measurements)

Charts show measured values in five-second aligned buckets. Host CPU is from the load-generator machine, not the target.






## Agent Activity Summary

| *Agent* | *Summary* |
|---|---|
| Test Execution | Load test executed on /comments?postId=1 (Unknown - Filtered Feedback API — Load Check).<br>Results: 718 total requests, 714 successful, 4 failed.<br>Avg response time: 239.3579094707519 ms; 95th percentile: 302.37209999999993 ms; Max: 1775.799 ms.<br>Throughput: 10.816712568144723 TPS; Error rate: 55.71%. |
| Monitoring | CPU usage: 9.85% average, 32.3% peak; memory: 78.99% average, 81.3% peak; DB connections: not collected, TPS: 10.816712568144723, Response time: 239.3579094707519 ms.<br>Data source and scope: REAL: psutil load-generator host samples and k6 client-observed metrics.<br>Measurement provenance: cpu_usage=psutil (load_generator_host); cpu_peak_usage=psutil (load_generator_host); memory_usage=psutil (load_generator_host); memory_peak_usage=psutil (load_generator_host); response_time=k6 (load_generator_observed_http_traffic); tps=k6 (load_generator_observed_http_traffic); db_connections=not_collected (No target-specific database telemetry provider is configured.).<br>Alerts triggered: ['High load-generator host memory usage'] |
| Security Validation | Security validation was not run for this job. |
| Analysis | This run of the Filtered Feedback API — Load Check scenario experienced a critical performance failure, with error rates and max response times dramatically breaching thresholds. Infrastructure monitoring recorded a warning, driven by high memory usage on the load generator host. No relevant security results or historical analysis context were available. Action is recommended on both error handling and infrastructure alerting. |
| Reporting | A PDF report was generated, including all findings, recommendations, and an executive summary. |

## Per-API Findings (Test Execution)

- /comments?postId=1 [FAIL / critical]
avg=239.3579094707519ms p95=302.37209999999993ms max=1775.799ms
throughput=10.816712568144723 req/s error_rate=55.71% http_status=200
* Error rate 55.71% exceeds 1.0% threshold.
* Max response time (1775.799ms) is 5.9x p95 (302.37209999999993ms) -- outlier spikes present.
history: No historical context was available; analysis is based on static threshold breaches alone.
recommendation: Address the error rate and response time outlier issues based on threshold findings. No matching SLA or incident documentation was found due to unavailable historical data.

## Infrastructure Health (Monitoring)

Status: WARNING (monitoring_agent reported: WARNING)
CPU usage: 9.85% Memory usage: 78.99% DB connections: None TPS: 10.816712568144723
Response time (monitoring): 239.3579094707519ms
- 1 monitoring alert(s) reported: ['High load-generator host memory usage']
Monitoring alerts: ['High load-generator host memory usage']

## Recommended Next Steps

- /comments?postId=1: Address the error rate and response time outlier issues based on threshold findings. No matching SLA or incident documentation was found due to unavailable historical data.
- Infrastructure: review CPU/memory/DB connection anomalies above.